

CO₂ system basics recap

Jack Middelburg



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Universiteit Utrecht

**Biogeochemical Processes
and Inorganic Carbon Dynamics**



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What you should know after this class

- Have your high school and/or bachelor chemistry refreshed 😊
- Basic understanding CO₂-H₂O system
- Alkalinity



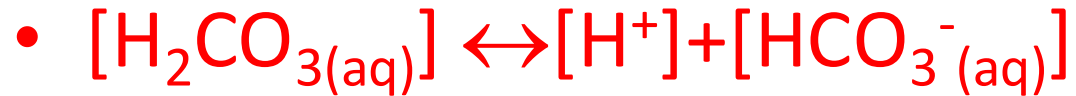
Consider H₂O

- $\text{H}_2\text{O} \leftrightarrow \text{H}^+ + \text{OH}^-$
- Equilibrium relation: $K_w = [\text{H}^+] [\text{OH}^-]$
- Solve
 - Two unknowns: $[\text{H}^+] [\text{OH}^-]$
 - One relation: $K_w = [\text{H}^+] [\text{OH}^-]$
 - One degree of freedom: one chosen, the other fixed

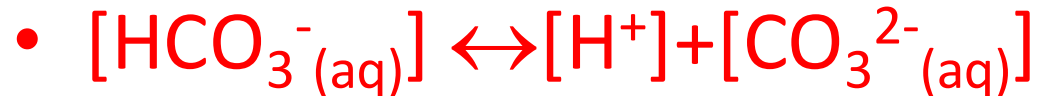
Consider CO₂-H₂O System



$$[\text{H}_2\text{CO}_{3(\text{aq})}] = K_{\text{H}} \cdot p\text{CO}_2$$



$$K_1 = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = 10^{-6.35}$$



$$K_2 = \frac{[\text{H}^+][\text{CO}_3^{2-}]}{[\text{HCO}_3^-]} = 10^{-10.33}$$

Dissolved CO_2 & H_2CO_3

- Dissolved $\text{CO}_{2(\text{aq})}$
- Gaseous $\text{CO}_{2(\text{g})}$
- Carbonic Acid $\text{H}_2\text{CO}_{3(\text{aq})}$
- $\text{H}_2\text{CO}_{3(\text{aq})} \ll \text{CO}_{2(\text{aq})}$
- $\text{CO}_{2(\text{aq})}$ and $\text{H}_2\text{CO}_{3(\text{aq})}$ cannot be analyzed independently
- They are therefore lumped and termed CO_2^* or H_2CO_3^*

How to solve?

- 5 aqueous species: H_2CO_3 , HCO_3^- , CO_3^{2-} , H^+ , OH^-
- 4 equilibria(mass action laws)

$$[H_2CO_{3(aq)}] = K_H \cdot pCO_2 \quad K_w = [H^+] [OH^-]$$

$$K_1 = \frac{[H^+][HCO_3^-]}{[H_2CO_3]} \quad K_2 = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]}$$

- One additional equation needed: two options

Two options for closure

1. **Charge balance:** positive and negative charge should balance:

$$[\text{H}^+] = [\text{OH}^-] + [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}]$$

2. **Proton balance:** recipe of water and carbonic acid:

$$[\text{H}^+] = [\text{H}^+]_{\text{H}_2\text{O}} + [\text{H}^+]_{\text{H}_2\text{CO}_3}$$

$$[\text{H}^+] = [\text{OH}^-] + [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}]$$

What can be measured?

- pH
- pCO₂
- Titration or total alkalinity (TA)
- Carbonate ion
- Dissolved Inorganic Carbon
 - DIC, C_T, TCO₂, ΣCO₂
 - $\text{DIC} = [\text{H}_2\text{CO}_{3(\text{aq})}] + [\text{HCO}_3^-_{(\text{aq})}] + [\text{CO}_3^{2-}_{(\text{aq})}]$

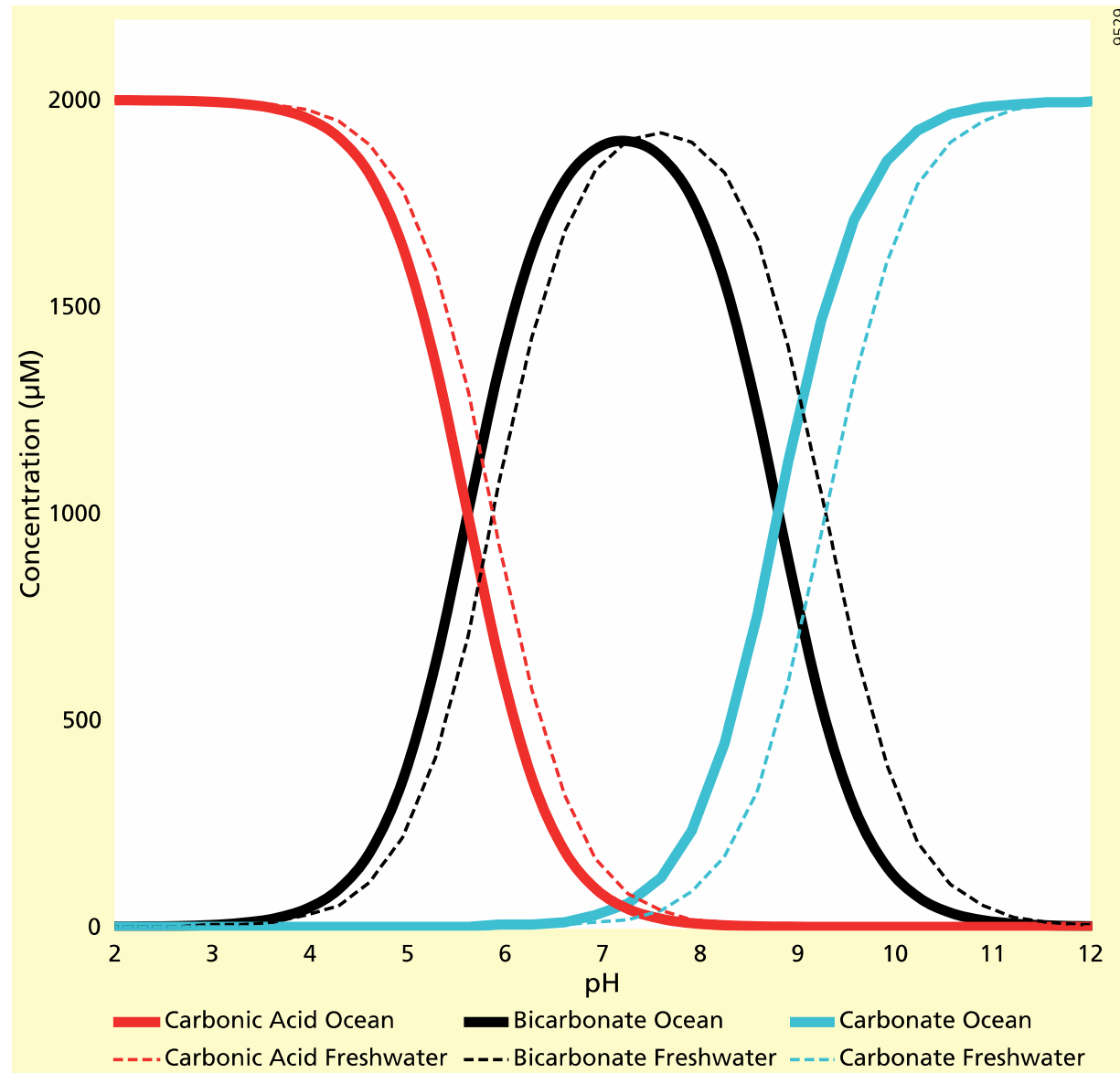
Total Alkalinity

- In words: Total alkalinity (TA) is the excess of proton acceptors over proton donors with respect to a zero level of protons
- Let us go back to the charge/proton balance:
 - $[H^+] = [OH^-] + [HCO_3^-] + 2[CO_3^{2-}]$
- Rearrange:
 - $TA = [OH^-] + [HCO_3^-] + 2[CO_3^{2-}] - [H^+]$
 - Exact definition of alkalinity for this system

Conservative properties

- When water is mixed then TA, DIC and salinity are conservative and set by mixing ratio
- pH, $p\text{CO}_2$, HCO_3^- etc are not conservative

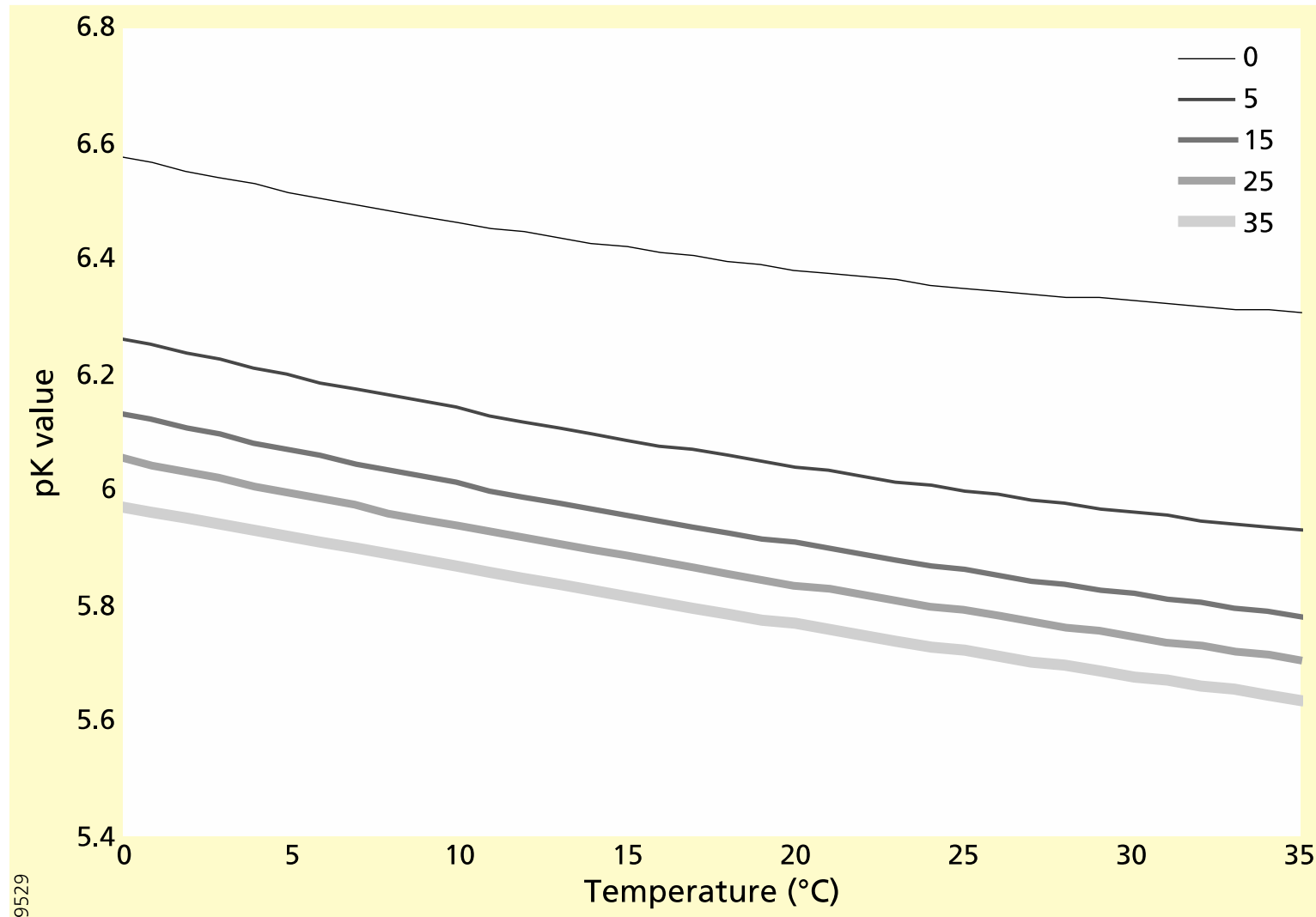
Bjerrum plot fresh and seawater



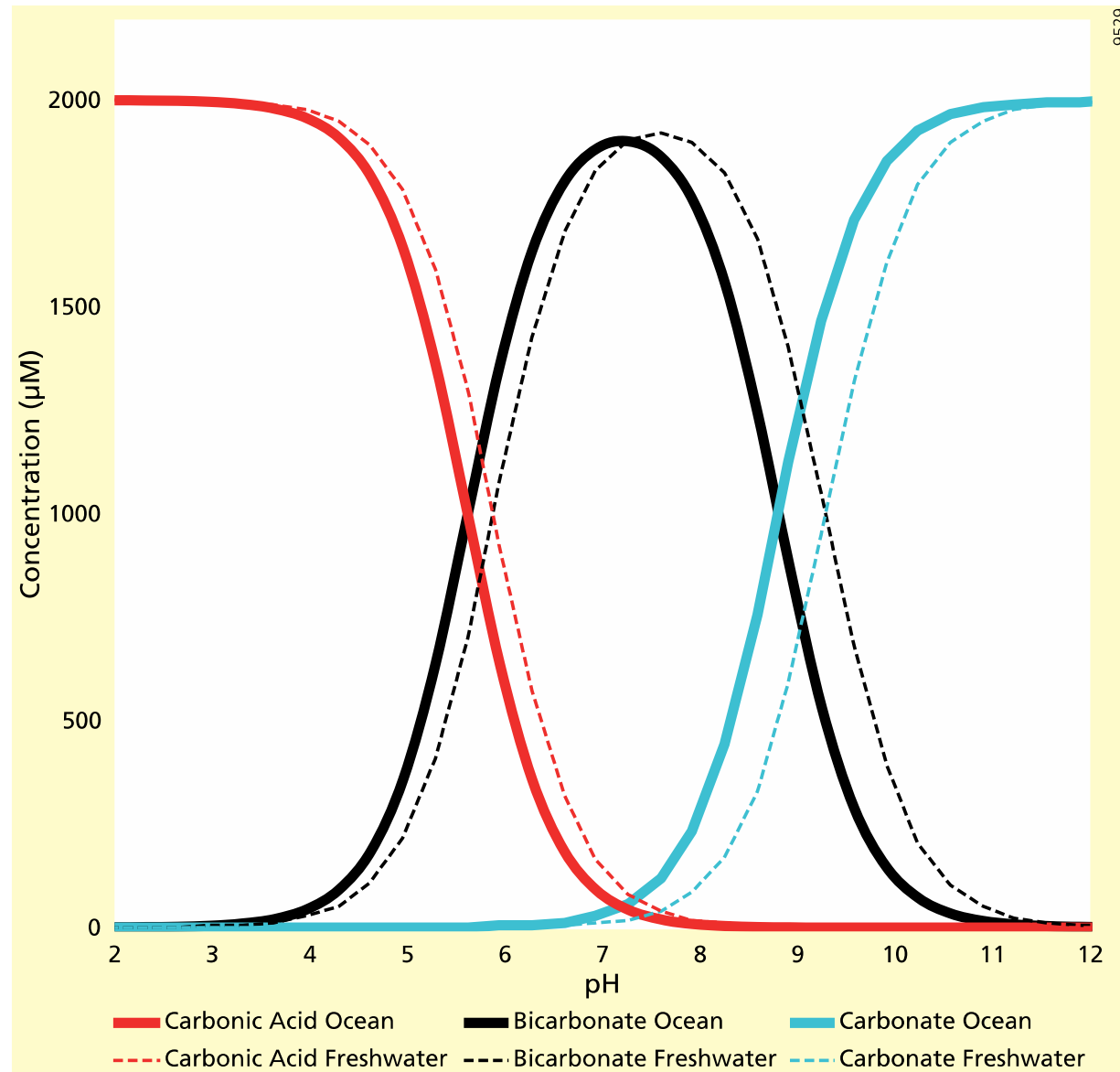
From Water to Seawater

- Ionic strength:
 - Non-linear effects
 - Concentration to activities
- Complexation with other ions in solution
- Two alternatives:
 - Include ionic strength and complexation corrections
 - Use equilibrium constants (apparent/stoichiometric) specific to seawater (K^*) that are a function of temperature, salinity (and pressure)
- Different pH scales

Stoichiometric constants depend on temperature and salinity



Bjerrum plot fresh and seawater

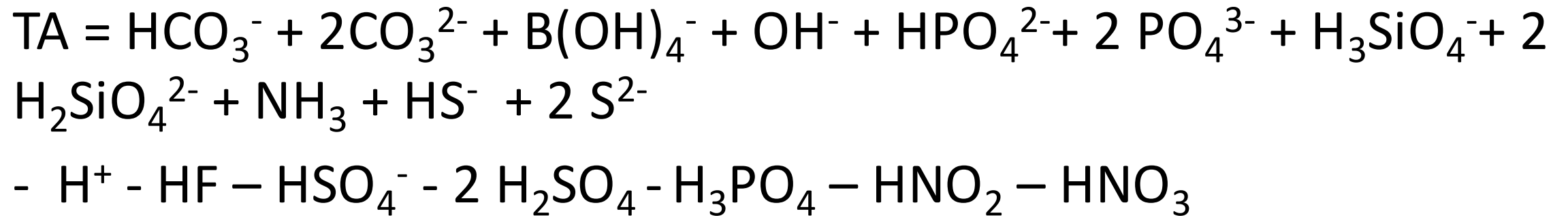


pH in seawater

- Total hydrogen ion scale: pH
- Other scales, not recommended, but often used:
 - NBS or IUPAC scale
 - SWS (seawater scale which includes HF and HSO_4^-)
 - Always use K^* with appropriate pH scale
- Conversions @ 25°C
 - $\text{pH} \approx \text{pH}_{\text{NBS}} - 0.15$ & $\text{pH} \approx \text{pH}_{\text{SWS}} + 0.01$

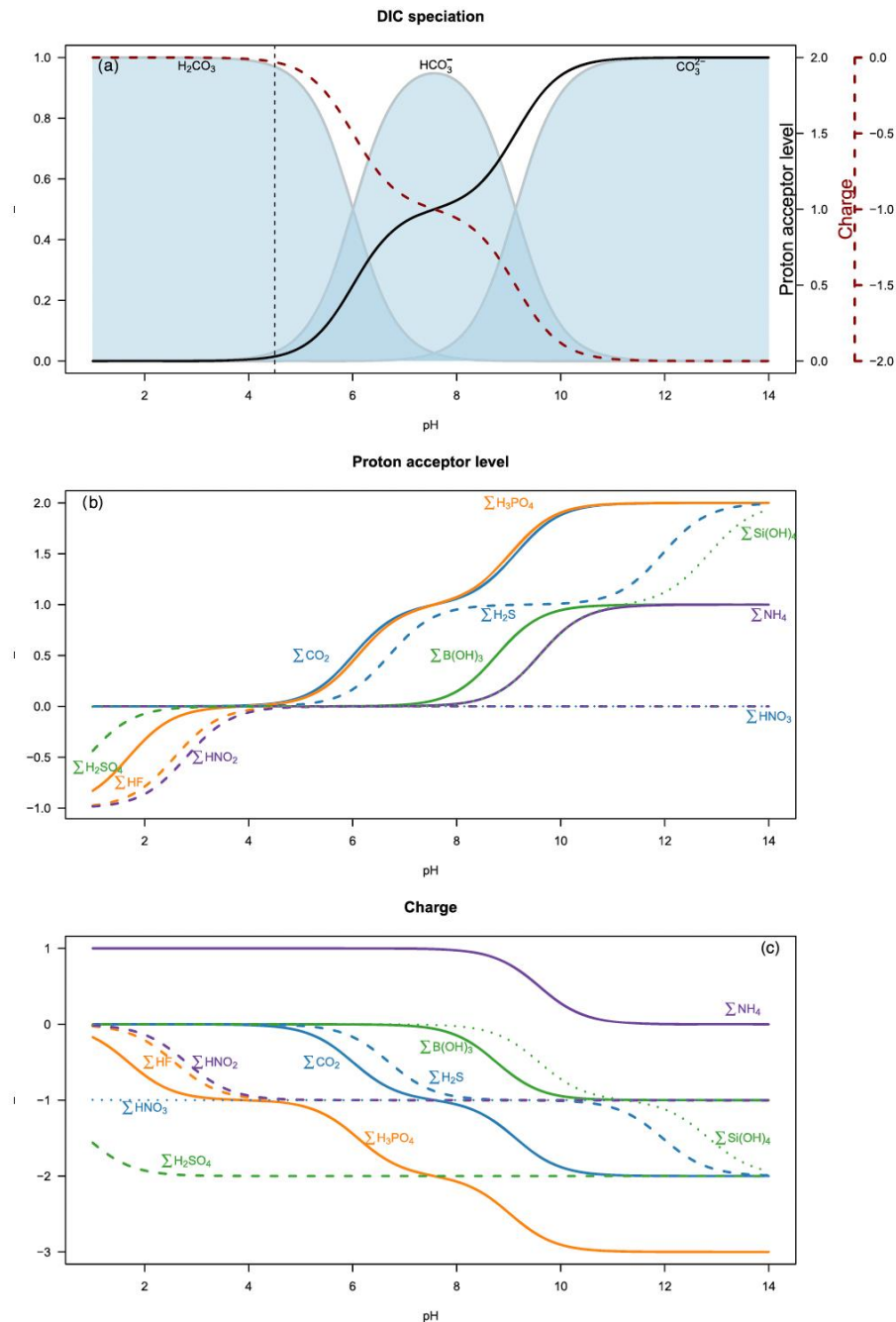
Alkalinity in Seawater

- More complex terms is needed: generally the Dickson definition is used:



- Carbonate alkalinity: $[\text{HCO}_3^-] + 2[\text{CO}_3^{2-}]$

Titration (proton) and Charge Balance Alkalinity



Dickson's definition and what is obtained by titration is based on a proton balance

Biogeochemical processes and calculations are based on charge-balance alkalinity

They are not exactly the same: there is difference because PO_4^{3-} , NO_3^- , NO_2^- , F^- , SO_4^{2-} that are charged at pH 4.5

Solving the system

- You always need two from the four measurable quantities to solve the system
 - pH and $p\text{CO}_2$
 - pH and DIC
 - pH and TA
 - Etc.

How to solve

- Some combinations can be solved analytically
- Usually this is done numerically:
 - CO2sys (Excell, Matlab, Phyton)
 - SWCO2
 - Seacarb in R
 - Aquaenv in R
 - CO2calc for iphone
 -

pH and DIC

$$[\text{CO}_2] = \text{DIC} \frac{[\text{H}^+]^2}{[\text{H}^+]^2 + [\text{H}^+] K_1^* + K_1^* K_2^*}$$

$$[\text{HCO}_3^-] = \text{DIC} \frac{[\text{H}^+] K_1^*}{[\text{H}^+]^2 + [\text{H}^+] K_1^* + K_1^* K_2^*}$$

$$[\text{CO}_3^{2-}] = \text{DIC} \frac{K_1^* K_2^*}{[\text{H}^+]^2 + [\text{H}^+] K_1^* + K_1^* K_2^*}$$

Global mean surface water speciation

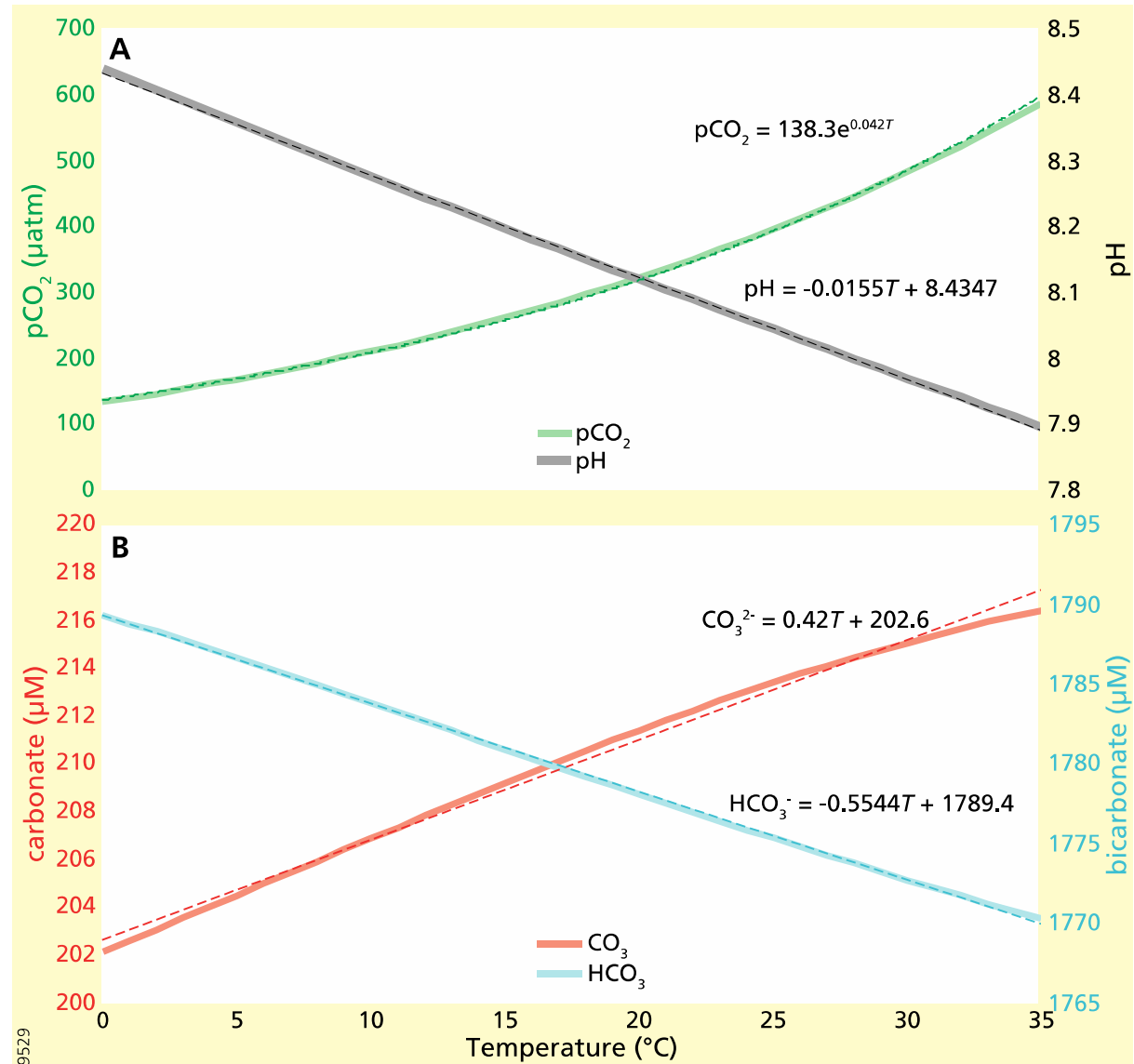
- $\text{DIC} = [\text{H}_2\text{CO}_{3(\text{aq})}] + [\text{HCO}_3^- (\text{aq})] + [\text{CO}_3^{2-} (\text{aq})]$
0.5% 88.6% 10.9%

- $\text{DIC} \approx [\text{HCO}_3^- (\text{aq})] + [\text{CO}_3^{2-} (\text{aq})]$

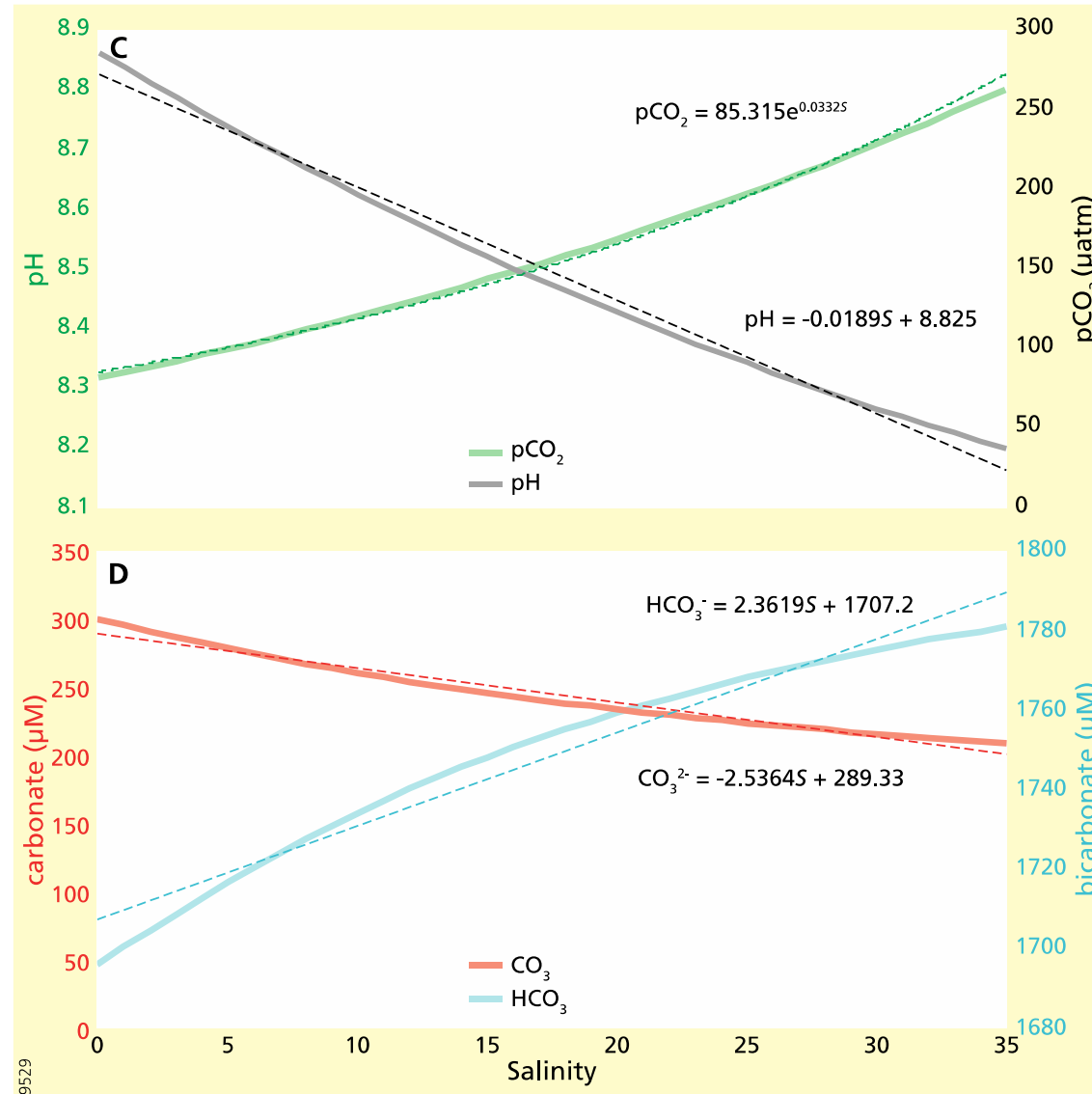
- $\text{TA} = [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}] + \text{others}$
76.8% 18.8% 4.4%

- $\text{Alkalinity} \approx \text{Carbonate alkalinity}$

Temperature, pH and pCO₂



Salinity, pH and pCO₂



Green and grey swapped, sorry